

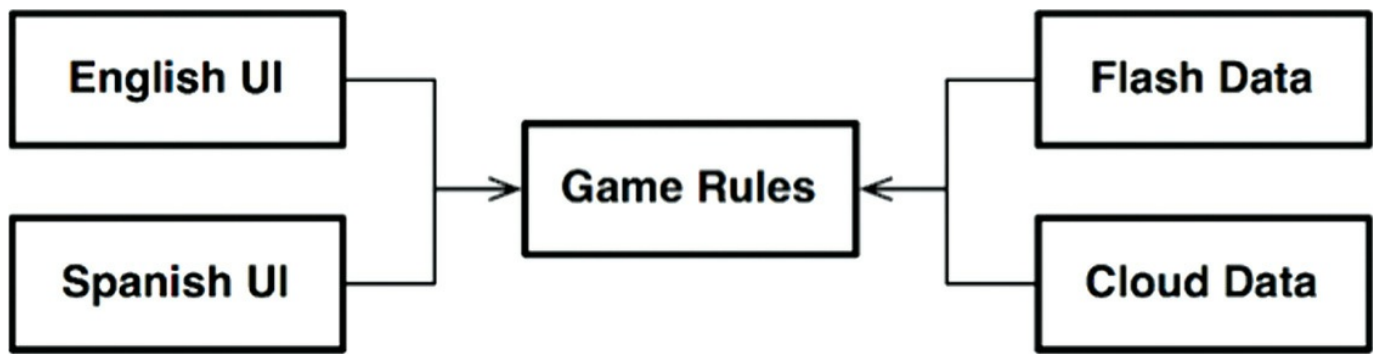


Figure 25.2 Following the Dependency Rule

CLEAN ARCHITECTURE?

It should be clear that we could easily apply the clean architecture approach in this context,¹ with all the use cases, boundaries, entities, and corresponding data structures. But have we really found all the significant architectural boundaries?

For example, language is not the only axis of change for the UI. We also might want to vary the mechanism by which we communicate the text. For example, we might want to use a normal shell window, or text messages, or a chat application. There are many different possibilities.

That means that there is a potential architectural boundary defined by this axis of change. Perhaps we should construct an API that crosses that boundary and isolates the language from the communications mechanism; that idea is illustrated in [Figure 25.3](#).

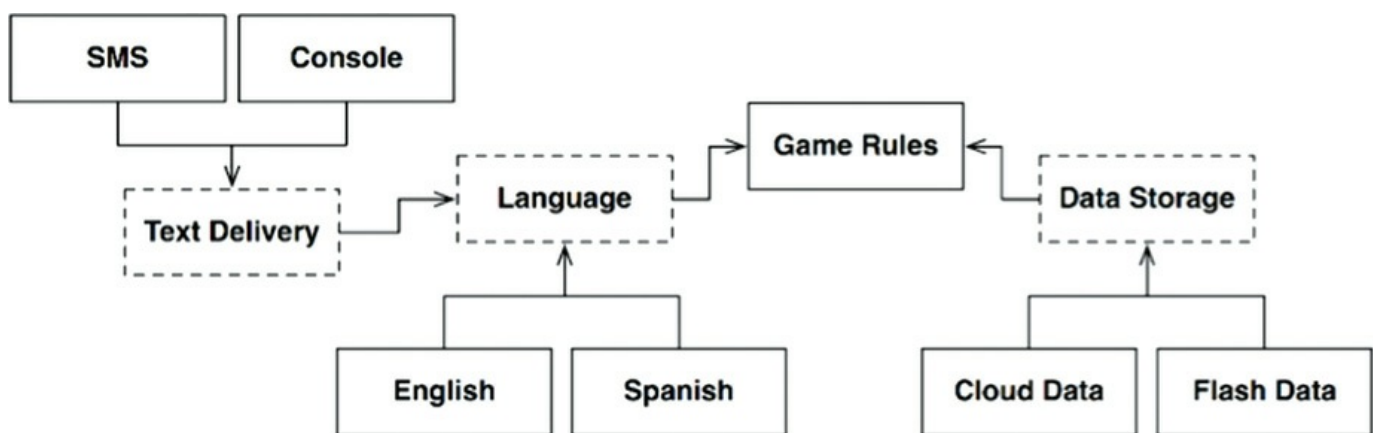


Figure 25.3 The revised diagram

The diagram in [Figure 25.3](#) has gotten a little complicated, but should contain no

surprises. The dashed outlines indicate abstract components that define an API that is implemented by the components above or below them. For example, the `Language` API is implemented by `English` and `Spanish`.

`GameRules` communicates with `Language` through an API that `GameRules` defines and `Language` implements. `Language` communicates with `TextDelivery` using an API that `Language` defines but `TextDelivery` implements. The API is defined and owned by the user, rather than by the implementer.

If we were to look inside `GameRules`, we would find polymorphic `Boundary` interfaces used by the code inside `GameRules` and implemented by the code inside the `Language` component. We would also find polymorphic `Boundary` interfaces used by `Language` and implemented by code inside `GameRules`.

If we were to look inside of `Language`, we would find the same thing: Polymorphic `Boundary` interfaces implemented by the code inside `TextDelivery`, and polymorphic `Boundary` interfaces used by `TextDelivery` and implemented by `Language`.

In each case, the API defined by those `Boundary` interfaces is owned by the upstream component.

The variations, such as `English`, `SMS`, and `CloudData`, are provided by polymorphic interfaces defined in the abstract API component, and implemented by the concrete components that serve them. For example, we would expect polymorphic interfaces defined in `Language` to be implemented by `English` and `Spanish`.

We can simplify this diagram by eliminating all the variations and focusing on just the API components. [Figure 25.4](#) shows this diagram.

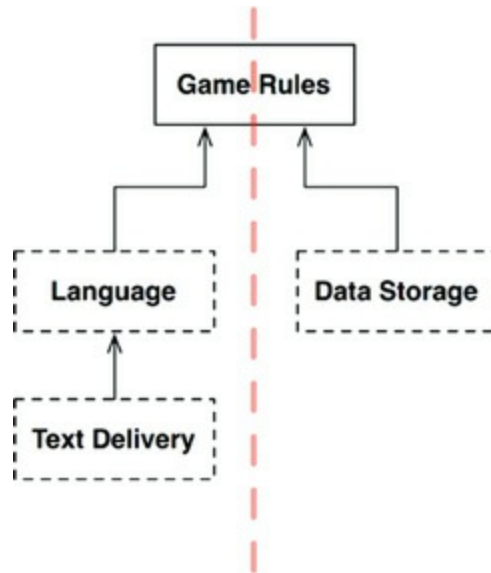


Figure 25.4 Simplified diagram

Notice that the diagram is oriented in [Figure 25.4](#) so that all the arrows point up. This puts `GameRules` at the top. This orientation makes sense because `GameRules` is the component that contains the highest-level policies.

Consider the direction of information flow. All input comes from the user through the `TextDelivery` component at the bottom left. That information rises through the `Language` component, getting translated into commands to `GameRules`. `GameRules` processes the user input and sends appropriate data down to `DataStorage` at the lower right.

`GameRules` then sends output back down to `Language`, which translates the API back to the appropriate language and then delivers that language to the user through `TextDelivery`.

This organization effectively divides the flow of data into two streams.² The stream on the left is concerned with communicating with the user, and the stream on the right is concerned with data persistence. Both streams meet at the top³ at `GameRules`, which is the ultimate processor of the data that goes through both streams.

CROSSING THE STREAMS

Are there always two data streams as in this example? No, not at all. Imagine that we would like to play Hunt the Wumpus on the net with multiple players. In this case, we would need a network component, like that shown in [Figure 25.5](#). This

organization divides the data flow into three streams, all controlled by the `GameRules`.

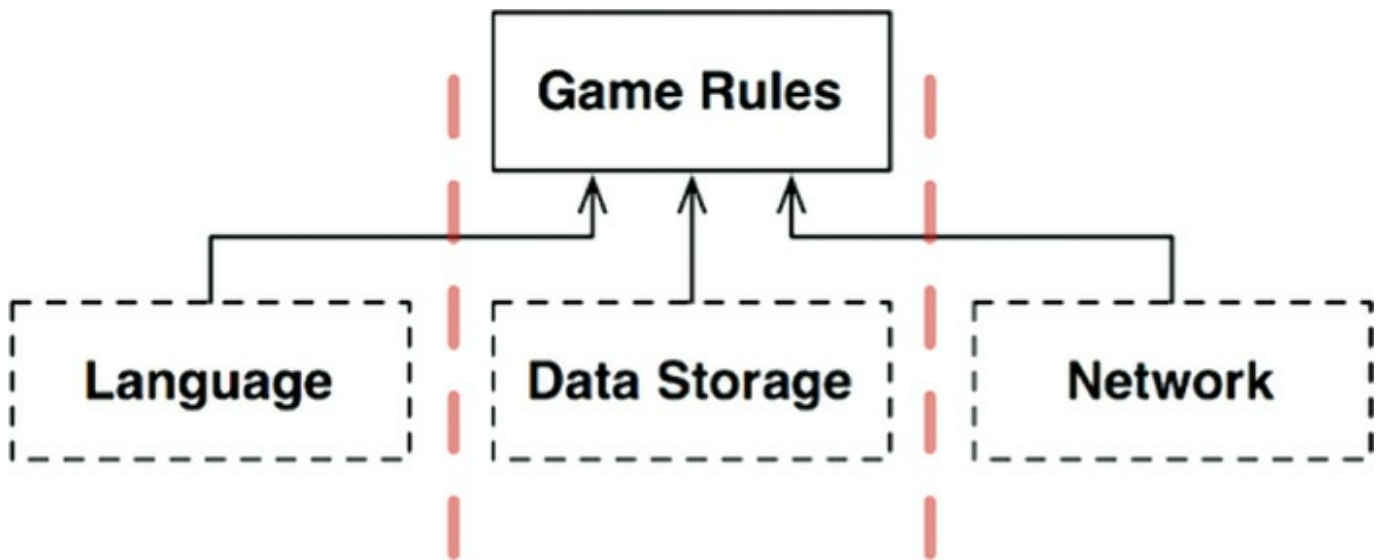


Figure 25.5 Adding a network component

So, as systems become more complex, the component structure may split into many such streams.

SPLITTING THE STREAMS

At this point you may be thinking that all the streams eventually meet at the top in a single component. If only life were so simple! The reality, of course, is much more complex.

Consider the `GameRules` component for Hunt the Wumpus. Part of the game rules deal with the mechanics of the map. They know how the caverns are connected, and which objects are located in each cavern. They know how to move the player from cavern to cavern, and how to determine the events that the player must deal with.

But there is another set of policies at an even higher level—policies that know the health of the player, and the cost or benefit of a particular event. These policies could cause the player to gradually lose health, or to gain health by discovering food. The lower-level mechanics policy would declare events to this higher-level policy, such as `FoundFood` or `FellInPit`. The higher-level policy would then manage the state of the player (as shown in [Figure 25.6](#)). Eventually that policy would decide whether the player wins or loses.

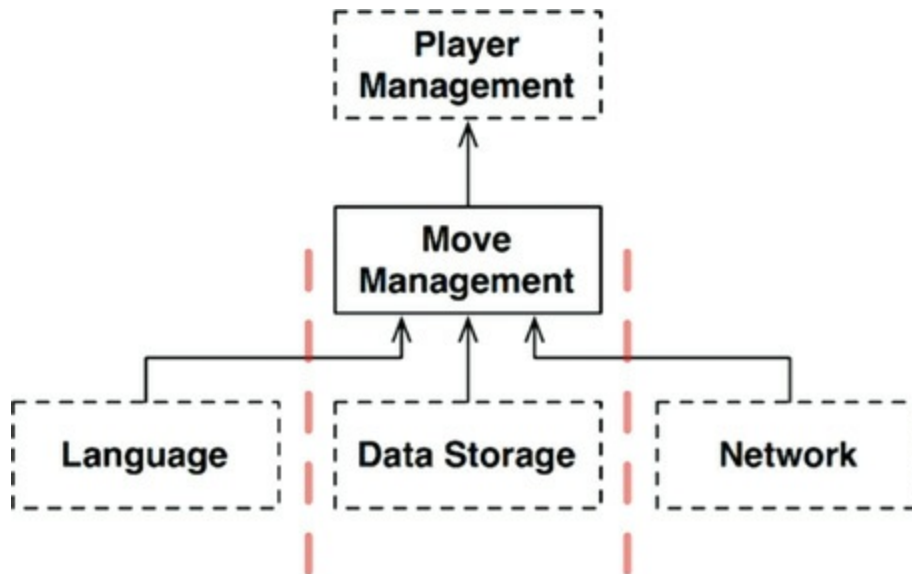


Figure 25.6 The higher-level policy manages the player

Is this an architectural boundary? Do we need an API that separates `MoveManagement` from `PlayerManagement`? Well, let's make this a bit more interesting and add micro-services.

Let's assume that we've got a massive multiplayer version of Hunt the Wumpus. `MoveManagement` is handled locally within the player's computer, but `PlayerManagement` is handled by a server. `PlayerManagement` offers a micro-service API to all the connected `MoveManagement` components.

The diagram in [Figure 25.7](#) depicts this scenario in a somewhat abbreviated fashion. The `Network` elements are a bit more complex than depicted—but you can probably still get the idea. A full-fledged architectural boundary exists between `MoveManagement` and `PlayerManagement` in this case.

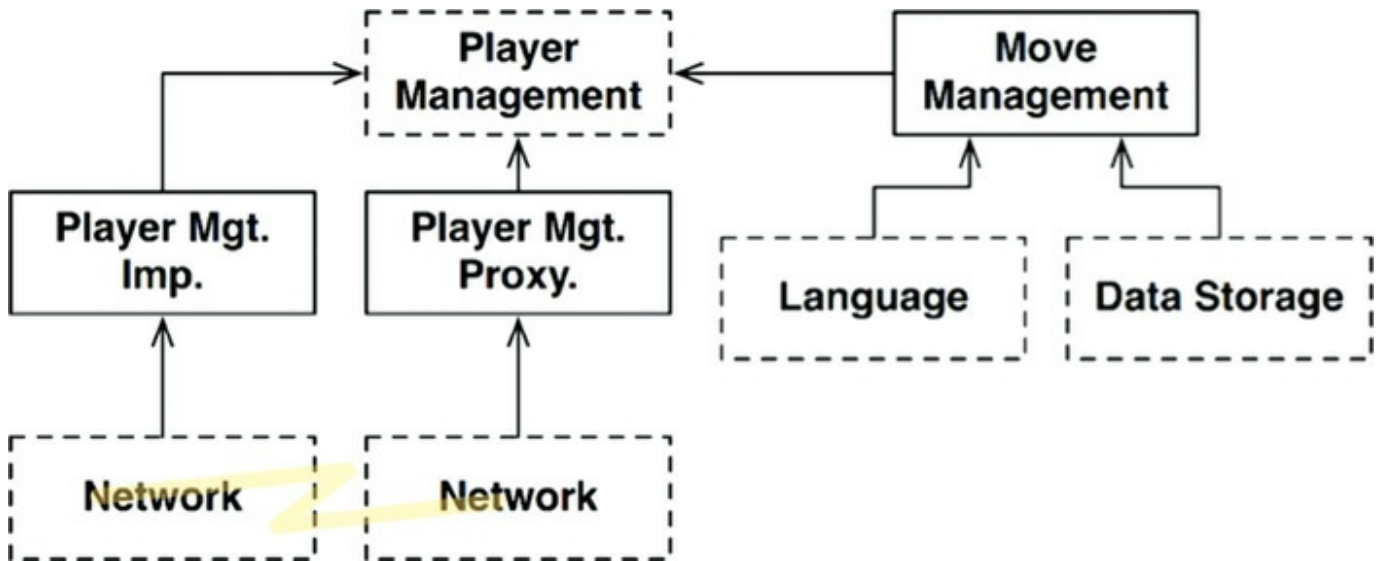


Figure 25.7 Adding a micro-service API

CONCLUSION

What does all this mean? Why have I taken this absurdly simple program, which could be implemented in 200 lines of Kornshell, and extrapolated it out with all these crazy architectural boundaries?

This example is intended to show that architectural boundaries exist everywhere. We, as architects, must be careful to recognize when they are needed. We also have to be aware that such boundaries, when fully implemented, are expensive. At the same time, we have to recognize that when such boundaries are ignored, they are very expensive to add in later—even in the presence of comprehensive test-suites and refactoring discipline.

So what do we do, we architects? The answer is dissatisfying. On the one hand, some very smart people have told us, over the years, that we should not anticipate the need for abstraction. This is the philosophy of YAGNI: “You aren’t going to need it.” There is wisdom in this message, since over-engineering is often much worse than under-engineering. On the other hand, when you discover that you truly do need an architectural boundary where none exists, the costs and risks can be very high to add such a boundary.

So there you have it. O Software Architect, you must see the future. You must guess—intelligently. You must weigh the costs and determine where the architectural boundaries lie, and which should be fully implemented, and which should be

partially implemented, and which should be ignored.

But this is not a one-time decision. You don't simply decide at the start of a project which boundaries to implement and which to ignore. Rather, you *watch*. You pay attention as the system evolves. You note where boundaries may be required, and then carefully watch for the first inkling of friction because those boundaries don't exist.

At that point, you weigh the costs of implementing those boundaries versus the cost of ignoring them—and you review that decision frequently. Your goal is to implement the boundaries right at the inflection point where the cost of implementing becomes less than the cost of ignoring.

It takes a watchful eye.

1. It should be just as clear that we would not apply the clean architecture approach to something as trivial as this game. After all, the entire program can probably be written in 200 lines of code or less. In this case, we're using a simple program as a proxy for a much larger system with significant architectural boundaries.
2. If you are confused by the direction of the arrows, remember that they point in the direction of source code dependencies, not in the direction of data flow.
3. In days long past, we would have called that top component the Central Transform. See *Practical Guide to Structured Systems Design*, 2nd ed., Meilir Page-Jones, 1988.

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THE MAIN COMPONENT



In every system, there is at least one component that creates, coordinates, and oversees the others. I call this component `Main`.

THE ULTIMATE DETAIL

The `Main` component is the ultimate detail—the lowest-level policy. It is the initial entry point of the system. Nothing, other than the operating system, depends on it. Its job is to create all the `Factories`, `Strategies`, and other global facilities, and then hand control over to the high-level abstract portions of the system.

It is in this `Main` component that dependencies should be injected by a `Dependency Injection` framework. Once they are injected into `Main`, `Main` should distribute those

dependencies normally, without using the framework.

Think of `Main` as the dirtiest of all the dirty components.

Consider the following `Main` component from a recent version of Hunt the Wumpus. Notice how it loads up all the strings that we don't want the main body of the code to know about.

[Click here to view code image](#)

```
public class Main implements HtwMessageReceiver {
    private static HuntTheWumpus game;
    private static int hitPoints = 10;
    private static final List<String> caverns = new ArrayList<>();
    private static final String[] environments = new String[]{
        "bright",
        "humid",
        "dry",
        "creepy",
        "ugly",
        "foggy",
        "hot",
        "cold",
        "drafty",
        "dreadful"
    };

    private static final String[] shapes = new String[] {
        "round",
        "square",
        "oval",
        "irregular",
        "long",
        "craggy",
        "rough",
        "tall",
        "narrow"
    };

    private static final String[] cavernTypes = new String[] {
        "cavern",
        "room",
        "chamber",
        "catacomb",
        "crevasse",
        "cell",
        "tunnel",
    };
```

```

    "passageway",
    "hall",
    "expanse"
};

private static final String[] adornments = new String[] {
    "smelling of sulfur",
    "with engravings on the walls",
    "with a bumpy floor",
    "",
    "littered with garbage",
    "spattered with guano",
    "with piles of Wumpus droppings",
    "with bones scattered around",
    "with a corpse on the floor",
    "that seems to vibrate",
    "that feels stuffy",
    "that fills you with dread"
};

```

Now here's the `main` function. Notice how it uses the `HtwFactory` to create the game. It passes in the name of the class, `htw.game.HuntTheWumpusFacade`, because that class is even dirtier than `Main`. This prevents changes in that class from causing `Main` to recompile/redeploy.

[Click here to view code image](#)

```

public static void main(String[] args) throws IOException {
    game = HtwFactory.makeGame("htw.game.HuntTheWumpusFacade",
                                new Main());

    createMap();
    BufferedReader br =
        new BufferedReader(new InputStreamReader(System.in));
    game.makeRestCommand().execute();
    while (true) {
        System.out.println(game.getPlayerCavern());
        System.out.println("Health: " + hitPoints + " arrows: " +
                            game.getQuiver());
        HuntTheWumpus.Command c = game.makeRestCommand();
        System.out.println(">");
        String command = br.readLine();
        if (command.equalsIgnoreCase("e"))
            c = game.makeMoveCommand(EAST);
        else if (command.equalsIgnoreCase("w"))
            c = game.makeMoveCommand(WEST);
        else if (command.equalsIgnoreCase("n"))

```